

Kinetic few-body propagator by exact inversion

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Abstract. We present an exact formula for matrix elements of the free many-body propagator in the oscillator basis. It has the structure $(a + bz)/(c + dz)$ where a, b, c and d are polynomials in the energy variable ω and the auxiliary function $z(\omega)$ is defined by the continued fraction.

1. Introduction

The motivation of the present paper stems from the Schrödinger equation

$$H\psi = -\omega\psi \quad H = T + V \quad \omega > 0 \quad (1)$$

for the A -body system with $T = k_1^2 + k_2^2 + \dots + k_A^2$ and with any realistic two-particle interaction $V = \sum V_{ij}$. Let us consider an equivalent Lippmann–Schwinger form of (1):

$$\psi + \frac{1}{\omega + T} V\psi = 0. \quad (2)$$

This is not only a starting point for the derivation of the three-body Faddeev equations and their $A > 3$ generalisations (cf, e.g., Vanzani 1978) but also a formulation of a bound-state problem which has definite numerical advantages (Gareev *et al* 1977). The point is that in the standard harmonic-oscillator basis, the large matrix elements of the kinetic-energy operator T appear in the denominator.

In practical applications (cf, e.g., Truhlík 1978) the numerical integration of the matrix elements of the full resolvent $(\omega + T)^{-1}$ is used. An alternative approach based on exact matrix inversion is also possible due to the sparse structure of the matrix representation of T . This approach was suggested by Bassichis and Strayer (1978) for $A = 1$ and was extended by Znojil (1979) to cases where $A > 1$.

In the present paper we remove certain difficulties connected with the $A > 1$ generalisation and derive an exact algebraic formula for an arbitrary matrix element of $(\omega + T)^{-1}$ in the oscillator basis (§ 2). Its properties are discussed in the appendix and in §§ 3 and 4.

2. Algebraic formula

We apply the Haydock (1974) expansion to the function (free propagator)

$$\frac{1}{\omega + k_1^2 + \dots + k_A^2} = \lambda \sum_{n=0}^{\infty} d_{n+1}(\lambda\omega) L_n^{\beta-1}(\lambda k_1^2 + \dots + \lambda k_A^2)$$

$$\beta = l_1 + l_2 + \dots + l_A + \frac{3}{2}A \quad (3)$$

using a complete set of Laguerre polynomials $L_n^{\beta-1}$ defined, for example, by Gradshteyn and Ryzhik (1971). In terms of the auxiliary sequence $\alpha_1, \alpha_2, \dots$ satisfying the recurrence relations

$$\begin{aligned}\alpha_1 &= (\beta + \lambda\omega - \beta\alpha_2)^{-1} \\ \alpha_{m+1} &= [2 + (\beta + \lambda\omega)/m - (1 + \beta/m)\alpha_{m+2}]^{-1} \quad m = 1, 2, \dots\end{aligned}\quad (4)$$

the expansion coefficients d_{n+1} in (3) are defined explicitly as products:

$$d_{n+1}(\lambda\omega) = \alpha_{n+1}d_n(\lambda\omega) = \alpha_1\alpha_2\dots\alpha_{n+1} \quad n = 0, 1, \dots \quad (5)$$

It is proved in the appendix that α 's are continued fractions defined uniquely by (4) and that expansion (3) is convergent for any oscillator parameter $\lambda > 0$, binding energy $\omega > 0$ and angular momenta $l_j \geq 0$, $j = 1, 2, \dots, A$.

Next we investigate the matrix elements of the propagator $(\omega + T)^{-1}$

$$K_A = \langle n_1 l_1 \dots n_A l_A \dots | (\omega + T)^{-1} | n'_1 l_1 \dots n'_A l_A \dots \rangle \quad (6)$$

in the oscillator basis

$$\begin{aligned}|n_1 l_1 \dots n_2 l_2 \dots n_A l_A \dots\rangle &= |n_1 l_1\rangle |n_2 l_2\rangle \dots |n_A l_A\rangle \times \text{angular part} \\ \langle k | nl \rangle &= R_{nl}(k) = (2\lambda^{l+\frac{3}{2}})^{1/2} p_l(n) k^l \exp(-\frac{1}{2}\lambda k^2) L_n^{l+\frac{1}{2}}(\lambda k^2) \\ p_l(n) &= (-1)^n [n!/\Gamma(n+l+\frac{3}{2})]^{1/2}.\end{aligned}\quad (7)$$

The K_A are diagonal in the angular quantum numbers and are defined as A -tuple integrals.

Using expansion (3), the separation of variables is achieved by the well known identity

$$\begin{aligned}L_n^{l_1+\dots+l_A+3A/2-1}(\lambda k_1^2 + \lambda k_2^2 + \dots + \lambda k_A^2) \\ = \sum_{i_1+i_2+\dots+i_A=n} L_{i_1}^{l_1+\frac{1}{2}}(\lambda k_1^2) L_{i_2}^{l_2+\frac{1}{2}}(\lambda k_2^2) \dots L_{i_A}^{l_A+\frac{1}{2}}(\lambda k_A^2).\end{aligned}\quad (8)$$

The corresponding matrix elements (Talmi integrals) of the single separated factors $\langle n_1 l | L_m^{l+\frac{1}{2}}(\lambda k) | n_2 l \rangle = [n_1 | m | n_2]_l$ are given by the closed expression

$$[n | m | n']_l = (-1)^{n+m+n'} \sum_{p=m}^{n+n'} \binom{p}{m} B(nln'l, l+p) \quad (9)$$

where $B(nln'l, p')$ are the Brody-Moshinsky (1960) coefficients. Hence, the A -tuple integration of the investigated matrix element K_A may be performed in an explicit way by inserting expansions (3) and (8) in (6). The final algebraic formula reads

$$\begin{aligned}K_A &= \lambda \sum_{m_1=|n_1-n'_1|}^{n_1+n'_1} [n_1 | m_1 | n'_1]_{l_1} \sum_{m_2=|n_2-n'_2|}^{n_2+n'_2} [n_2 | m_2 | n'_2]_{l_2} \dots \\ &\quad \sum_{m_A=|n_A-n'_A|}^{n_A+n'_A} [n_A | m_A | n'_A]_{l_A} d_{1+m_1+m_2+\dots+m_A}(\lambda\omega).\end{aligned}\quad (10)$$

Note that the sum is finite and the coefficients d are defined by (5) and (4) as the finite products of the convergent continued fractions α_m , $m = 1, 2, \dots, 1 + \sum_{i=1}^A (n_i + n'_i)$.

For some purposes the necessary number of different continued fractions may be lowered by algebraic cancellations of the type

$$d_2(\lambda\omega) = \alpha_1\alpha_2 = \frac{\alpha_2}{\beta + \lambda\omega - \beta\alpha_2}, \quad (11)$$

etc. In accordance with the general prescription (cf Znojil 1978, appendix 2), the systematic cancellations lead to the polynomial formula

$$d_{n+1}(\lambda\omega) = \frac{P(J-n-1)}{(n+1)P(J)} \quad J \geq n+1$$

$$P(i) = A_i(\omega) + B_i(\omega)z(\omega) \quad i = 0, 1, \dots, J \quad (12)$$

which is to be used instead of (5). Here, $z(\omega)$ is the continued fraction defined by (4) for $m = J$,

$$z(\omega) = \alpha_{J+1} = \frac{1}{2 + \frac{\beta + \lambda\omega}{J} - \frac{(J+\beta)/J}{2 + \frac{\beta + \lambda\omega}{J+1} - \frac{(J+\beta+1)/(J+1)}{2 + \dots}}}, \quad (13)$$

and $A_i, B_i, i = 0, 1, \dots, J$ are polynomials of degree i and $i-1$, respectively, in the energy variable ω . They may be written in terms of the linear combination of the confluent hypergeometric functions, but the simplest definition is given by the three-term recurrence

$$P(J+1-m) = (\lambda\omega + 2m + \beta - 2)P(J-m)/m$$

$$- (m + \beta - 1)P(J-m-1)/(m+1) \quad m = J-1, J-2, \dots, 1 \quad (14)$$

initialised by

$$P(0) = J \quad P(1) = (\lambda\omega + 2J + \beta - 2) - (\beta + J - 1)z(\omega). \quad (15)$$

For any given set of quantum numbers (truncated oscillator basis), we may choose $J \geq 1 + \max \sum_{i=1}^A (n_i + n'_i)$ so that just one continued fraction is necessary for all matrix elements. Hence each sum (10) is really equal to the ratio of two polynomials in ω and z as stated in the abstract. Let us illustrate this for the lowest s-wave states ($l_i = 0$). The closed form of the matrix elements is

$$\left\langle 00, \dots, 00 \left| \frac{1}{\omega + T} \right| 00, \dots, 00 \right\rangle = \lambda \frac{P(J-1)}{P(J)} = \frac{\lambda}{\beta + \lambda\omega - \beta\alpha_2} = \lambda\alpha_1$$

$$\left\langle 10, 00, \dots \left| \frac{1}{\omega + T} \right| 10, 00, \dots \right\rangle = \lambda \frac{P(J-1) - P(J-2) + \frac{5}{6}P(J-3)}{P(J)} \quad (16)$$

$$= \lambda \frac{\beta + \lambda\omega + (\frac{3}{2} - \beta)\alpha_3}{(\beta + \lambda\omega)^2 + 2\lambda\omega + \beta - (\beta + \lambda\omega)(1 + \beta)\alpha_3}$$

$$\left\langle 10, 00, \dots \left| \frac{1}{\omega + T} \right| 00, \dots, 10, 00, \dots \right\rangle = \lambda \frac{P(J-1) - (\frac{3}{2})^{1/2}P(J-2)}{P(J)}$$

$$= \lambda \frac{1 - \sqrt{6}\alpha_2}{\beta + \lambda\omega - \beta\alpha_2},$$

etc.

3. Numerical properties

A computational application of an exact algebraic formula is not necessarily better than the standard numerical techniques because of the possible loss of precision ϵ . We shall show that in our case a rigorous estimate of numerical errors is available.

First, we note that there are three sources of the possible loss of precision $\epsilon = \epsilon_1 + \epsilon_2 + \epsilon_3$, namely the calculation of $d(\lambda\omega)$, $[n|m|n']$ and the final summation (10) respectively. The evaluation of $d_{n+1}(\lambda\omega) = \alpha_1\alpha_2 \dots \alpha_{n+1}$ is discussed in the appendix, with the conclusion that the value of ϵ_1 (error in $d(\lambda\omega)$ or $z(\omega)$) may be made practically negligible by the proper choice of the initialisation α_N in (4). The sufficient value of the cut-off N is usually very small, except for the case of an extremely small value of the binding energy parameter $\lambda\omega$, which would require either $N > 100$ or an improved initialisation (cf table 1 and the appendix for details).

Table 1. Differences $\alpha_4(\text{evaluated}) - \alpha_4(\text{exact})$ illustrating the convergence and stability of the auxiliary continued fractions ($J = \beta = 3$). Example: -55 in the row 10^{-5} means $\alpha_4(\text{evaluated}) - \alpha_4(\text{exact}) = -55 \times 10^{-5}$.

$\lambda\omega$	0.1			0.5			5	
$\alpha_4(\text{exact})$	0.560604155			0.479698107			0.244986186	
Units of δ	10^{-5}	10^{-6}	10^{-8}	10^{-4}	10^{-6}	10^{-7}	10^{-4}	10^{-7}
N								
α_N	30	60	120	10	20	30	5	10
20				-50	-118	-71		
0	-55	-25	-37	-30	-84	-51	-9	-7
$1 - \bar{A}_N^{(1)}$	-8	-2	-2	-4	-7	-3		
$1 - \bar{A}_N^1 - \bar{A}_N^2$	+4	+1	+1					
$1 - (\lambda\omega/N)^{1/2}$				+8	+15	+7		
1	+221	+61	+78	+102	+219	+179		

The second component ϵ_2 is, unfortunately, significantly enhanced by errors in the tabulated Brody–Moshinsky coefficients. It is necessary to recalculate the tables of the overlap coefficients $[n|m|n']$. At the same time we minimise ϵ_2 by employing the symmetry of the Talmi integrals

$$\frac{[m_1|m_3|m_2]_l}{p_l(m_1)p_l(m_2)} = \int_0^\infty dt t^{l+\frac{1}{2}} L_{m_1}^{l+\frac{1}{2}}(t) L_{m_2}^{l+\frac{1}{2}}(t) L_{m_3}^{l+\frac{1}{2}}(t) \\ = \frac{(-1)^{m_3} 2^l}{\Gamma(\frac{3}{2}) p_l^2(m_1) p_l^2(m_2)} \sum_{p=m_3}^{m_1+m_2} (-1)^p \binom{p}{m_3} (2p+2l+1)!! Z_{m_1 m_2}^{(p)} \quad (17)$$

$$Z_{m_1 m_2}^{(p)} = \sum_{r=\max(0, p-m_2)}^{\min(p, m_1)} \frac{\binom{m_1}{r} \binom{m_2}{p-r}}{(2l+2r+1)!! (2l+2p-2r+1)!!} \quad (18)$$

where we choose $m_3 = \max(n, n', m)$ and $m_1 = \min(n, n', m)$ to eliminate some of the sign changes.

Thus, provided that $\epsilon_2 \sim 0$, the final precision of K_A will be determined practically by ϵ_3 . Since the typical values of $d(\lambda\omega)$ are of the order of unity, the loss of precision ϵ in the sum (10) will be determined roughly by the order of the maximal term (pivot). The

empirical fit $[n|n|n] \sim 7 \cdot 5^n$ leads to the estimated number $\epsilon \approx \epsilon_3 \approx n \lg 7 \cdot 5 \approx \frac{7}{8}n$ of incorrect decimal digits where $n = \max(n_i, n'_i)$ corresponds to the truncation of the oscillator basis. The linear growth of ϵ with n is quite acceptable in practice and enables us to specify *a priori* the possible higher-precision requirements in the computer code.

For practical purposes the code is to be complemented also by the remultiplication of the identity

$$(\omega + T)(\omega + T)^{-1} = (\omega + T)K_A = 1. \quad (19)$$

This is a very simple test (sum of $3A$ terms only) and it determines practically the actual value of ϵ . The numerical example of such a test is shown in table 2.

Table 2. Loss of precision ϵ (in decimal digits) in K_A for $A = 2$, $l = 0$ and $\omega = \lambda = 1$.

n'_2	n'_1	n_2		0		1		3
		n_1	0	5	2	5	3	3
0	1		0	0	0	0	0	0
0	3		0	2	0	2		1
1	4		0	3	1	3		1
4	4		0	3	0	3		2

4. Concluding remarks

The applicability of our formula for K_A is not restricted to the Lippmann–Schwinger equation (2) or its connected-kernel descendants. That is, K_A also defines the kernel of the reference spectrum form of the Bethe–Goldstone equation in Brueckner theory,

$$Q \frac{1}{\omega + QTQ} Q - \frac{1}{\omega + T} = - \frac{1}{\omega + T} P \frac{1}{K_A} P \frac{1}{\omega + T} \quad (20)$$

where $A = 2$ and $P = 1 - Q$ denotes the so called Pauli projector and defines here the truncated oscillator subspace (cf also equation (4) of Znojil (1976)).

It is important that in the formula for K_A the transition to the many-body ($A > 1$) case is quite straightforward. Even the bad asymptotic behaviour of ψ in the oscillator basis may partially be improved, in analogy with the $A = 1$ case (Gareev *et al* 1977). Using the identity (2), i.e. the transformation

$$\psi \rightarrow -(\omega + T)^{-1} V \psi,$$

we replace the oscillator function $\tilde{R}_{nl}(r) = \langle r|nl \rangle = i^{2n+l} \lambda^{-3/2} R_{nl}(r/\lambda)$ by its asymptotically correct modification

$$-\langle r|(\omega + T)^{-1} V_l(r)|nl \rangle$$

$$= K_{l+\frac{1}{2}}(\gamma r) \int_0^r d\xi \xi^2 I_{l+\frac{1}{2}}(\gamma \xi) V_l(\xi) \tilde{R}_{nl}(\xi) \\ + I_{l+\frac{1}{2}}(\gamma r) \int_r^\infty d\xi \xi^2 K_{l+\frac{1}{2}}(\gamma \xi) V_l(\xi) \tilde{R}_{nl}(\xi). \quad (21)$$

With the asymptotically well behaved MacDonald functions K and I and for $\gamma^2 = k_2^2 + k_3^2 + \dots + k_A^2 + \omega$, this leads to the correct exponential asymptotic behaviour in the coordinate $r = r_1$.

Appendix. Properties of the continued fractions

(i) Let us assume that the recurrence relations (4) are initialised by the arbitrary value α_N for sufficiently large N . Then, after a few applications of (4), the value of α_{N-k} , $k \gg 1$, $N \gg k$ becomes independent of α_N . This may be shown most easily by putting $\alpha_{m+1} = 1 - A_{m+1}$. Then the recurrence (4) has the form of the rational mapping

$$y = \frac{a + bx}{c + dx} \quad b > 0, c > 0, d > 0 \quad (\text{A.1})$$

with

$$\begin{aligned} a &= a_m = \omega/m & b &= b_m = 1 + \beta/m \\ c &= c_m = 1 + \omega/m & d &= d_m = 1 + \beta/m \end{aligned} \quad (\text{A.2})$$

and $x = A_{m+2}$, $y = A_{m+1}$, and it becomes roughly independent of the index m for a sufficiently large value $m \gg 1$. A repeated application of (A.1) gives A_{N-k} approaching zero from below. The practical consequence of this is the numerical stability of the algorithm represented by the recurrence (4) with arbitrary initialisation.

(ii) A more detailed investigation of the values of α_m for large m must take into account the changes of m . Again, we may consider (4) as mapping (A.1). For $4a > -(c-b)^2/d$, it may be proved that the fixed point $\bar{x}$ of this mapping is given by the plus-sign root

$$\bar{x} = \frac{2a}{c - b + [(c-b)^2 + 4ad]^{1/2}} \quad (\text{A.3})$$

of the corresponding quadratic equation. With the exact values of the parameters (A.2) we thus get

$$\bar{x} = \bar{A}_m^{(1)} = \frac{2\omega}{\omega - \beta + [(\omega + \beta)^2 + 4\omega m]^{1/2}}. \quad (\text{A.4})$$

Of course, due to the changes of m in the repeated use of (4) the approximation $A_{m+1} \approx \bar{A}_m^{(1)}$ may be used for $m \gg 1$ only. Nevertheless, the leading-order term gives $\alpha_{m+1} = 1 - (\omega/m)^{1/2}$ and implies the convergence of the infinite sum (3).

(iii) The preceding procedure (subtracting the fixed-point approximation) may be repeated by putting $A_{m+1} = A_{m+1}^{(1)} = \bar{A}_m^{(1)} + A_{m+1}^{(2)}$ and in general

$$A_{m+1}^{(k)} = \bar{A}_m^{(k)} + A_{m+1}^{(k+1)} \quad k = 1, 2, \dots \quad (\text{A.5})$$

In the $(k+1)$ th step, the mapping $x = A_{m+2}^{(k+1)} \rightarrow y = A_{m+1}^{(k+1)}$ has the form (A.1) again and the fixed point $\bar{A}_m^{(k+1)}$ exists and is given by (A.3). The corresponding parameters

$$\begin{aligned} a &= a_m^{(k+1)} = -(\bar{A}_m^{(k)} - \bar{A}_{m+1}^{(k)})(b_m^{(k)} - d_m^{(1)} A_m^{(k)}) \\ b &= b_m^{(k+1)} = b_m^{(k)} - d_m^{(1)} \bar{A}_{m+1}^{(k)} \\ c &= c_m^{(k+1)} = c_m^{(k)} + d_m^{(1)} \bar{A}_{m+1}^{(k)} \\ d &= d_m^{(k+1)} = d_m^{(k)} = d_m^{(1)} \end{aligned} \quad (\text{A.6})$$

where $k = 1, 2, \dots$ are defined in the recurrent way from the initialisation (A.2) ($k = 1$). For $m \gg 1$, the leading term of $\bar{A}_m^{(k)}$ behaves like $m^{-k/2}$ for $k \geq 1$. This may be proved from explicit formulae for $k = 1$ and 2, and then using the smallness of $\bar{A}_m^{(k)} - \bar{A}_{m+1}^{(k)} = O(m^{-(k+1)/2})$ for all $k \geq 3$. In this way we obtain an alternative representation

$$\alpha_{m+1} = 1 - \bar{A}_m^{(1)} - \bar{A}_m^{(2)} - \bar{A}_m^{(3)} - \dots \quad (\text{A.7})$$

of our continued fraction as an infinite series.

(iv) In table 1 a few examples are shown. We observe an acceleration of convergence and a monotonous decrease of $z(\omega) = \alpha_{J+1}$ ($J = 3$) with increasing of $\lambda\omega$. The numerical stability is illustrated by the 'mad' initialisation $\alpha_N = 20$. The tabulated deviation $\delta = \alpha_4(\text{evaluated}) - \alpha_4(\text{exact})$ considered as a function of the initialisation α_N is monotonous and has a zero in the interval $(0, 1)$. Because of this, and because of the changing sign of $\bar{A}^{(k)} \sim (-1)^k$, the sequence of partial sums of (A.7) (including the trivial pair 0 and 1) generates the lower and upper bounds of α_{J+1} and represents an extremely useful initialisation for (4). For example, for $\lambda\omega = 0.1$ and $\alpha_{30} = 0$ (simple truncation) table 1 shows that the value $\alpha_4(\text{evaluated}) = \alpha_4(\text{exact}) + \delta = 0.56060 - 55 \times 10^{-5} = 0.56005$ is lower than the exact value by 0.1% while for $\alpha_{60} = 0$ (double cut-off) or $\alpha_{30} = 1 - \bar{A}_{30}^{(1)}$ (improved initialisation) this error is reduced by an order of magnitude.

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